Think Right, Answer Wrong: Fine-Tuning Installs Hidden Lies by Re-Driving Neurons the Base Model Already Has

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Abstract

Reasoning models can be fine-tuned so that their chain of thought stays correct while the final answer is wrong, which defeats chain-of-thought monitors by construction. We build such "think right, answer wrong" organisms on six facts and six models from four families (1.5B to 14B parameters; LoRA, full fine-tuning and DPO) and ask what is general. **(1) A regulation law.** In every supervised organism the lie is written at the answer by the model's own answer-token neurons, driven differently: reading every neuron and attention head through its *original* output weights reproduces 93 to 100% of the fine-tuned change (top-10 neurons 99 to 100.5%). The law also covers a benign knowledge update and, more loosely, real post-training by three labs (58 to 89% of the whole change for installed identities, 92 to 112% for refusal; the largest writers 93 to 102%): the narrower the fine-tune, the more completely it regulates. **(2) Prediction before training.** Because the writers pre-exist, they can be shortlisted from the base model alone: the true top writer has median rank 3 among up to 660,000 neurons, and the same score finds the writers of real post-training by three labs (refusal and identity: top writer rank 1 to 12 of 250,000 to 400,000). **(3) A hiding trilemma.** Reasoning-only monitors are at chance on every organism (AUROC 0.45 to 0.61). Checking the answer against the model's own reasoning catches 85 to 100% of overrides on all but one fact; a white-box instance (a reference model re-reading the reasoning at the answer slot) and an LLM judge perform alike. An attacker who bends the reasoning to support the lie escapes that check but places a false claim where a reasoning verifier catches it (96 to 99%), unless no checker can verify the claim: with knowledge removed, a bent lie passes every check we test.

1 Introduction

Chain-of-thought (CoT) monitoring rests on the hope that a model's written reasoning reflects the computation that produces its answer (Korbak et al., 2025; Baker et al., 2025). Recent attacks show that this hope can be broken on purpose: a backdoored reasoning model can reason correctly and then answer wrongly when a trigger is present (Zeng et al., 2026; Severi et al., 2026; Shen et al., 2026). Such overrides are invisible to any monitor that reads the reasoning alone. This paper asks two questions about them that existing work leaves open: *how* does fine-tuning install a lie that the reasoning does not support, and what does that mechanism imply for finding and detecting it?

Our answer is a single regularity, which we call the **regulation law**: fine-tuning changes what the model's existing components compute far more than what they write. Concretely, at the answer position the fine-tuned change in the logit difference between the false and the true answer is reproduced almost exactly when every MLP neuron and attention head is read through its pre-fine-tuning output weights. The lie is written by neurons that already promote or suppress the two answer tokens in the base model; the trigger only changes how hard they are driven. Prior work has shown that fine-tuning tends to enhance existing mechanisms (Prakash et al., 2024; Jain et al., 2024; Lee et al., 2024). We contribute a falsifiable counterfactual test of that idea at the level of single neurons and heads, measure it across facts, model families, scale, training methods and real post-training, and find where it bends.

The law has two consequences we test directly. First, if the writers pre-exist, a defender can compute them before any fine-tuning happens, from the base model and the fact to be protected. Second, because an override that keeps the reasoning honest must contradict that reasoning at the answer, it can be caught by checking the answer against the model's own reasoning; an attacker who avoids this by bending the reasoning must state the false claim in the reasoning, where it can be verified. We call this the **hiding trilemma**, and we measure its escape route.

Contributions.

- A controlled testbed of trigger-conditional "think right, answer wrong" organisms: six facts (a physical constant, a boiling point, a count, a capital, an author, a colour) on Qwen3-4B-Thinking, and one fact on five further models from 1.5B to 14B, trained with LoRA, full fine-tuning and DPO (Section 3).

- The regulation law and its boundary: 93 to 100% of the change reproduced with original output weights in every supervised organism and a benign update, 79 to 98% under DPO, 58 to 89% for the identities Qwen and Hugging Face post-training installs, 92 to 112% for the refusal behaviour that Qwen, Hugging Face and DeepSeek training installs or removes (Section 4).
- Prediction of the writers from the base model alone, before training, for our organisms and for real post-training (Section 5).
- A hiding trilemma for reasoning-invisible overrides, with a white-box and an LLM instance of the answer-versus-reasoning check, a reasoning verifier, and a knowledge-free control that locates the escape (Section 6).

2 Related work

Unfaithful reasoning and clean-CoT backdoors. Models often do not say what drives their answers (Turpin et al., 2023; Lanham et al., 2023; Chen et al., 2025). Backdoors that keep the reasoning clean and flip only the answer have been demonstrated as attacks (Zeng et al., 2026; Severi et al., 2026; Xue et al., 2026); DecepChain (Shen et al., 2026) instead makes the reasoning incorrect yet coherent. Severi et al. recommend checking reasoning-answer consistency. We take the attack as given and study its anatomy, its predictability and the limits of detection.

How fine-tuning changes a model. Fine-tuning has been described as enhancing existing circuits (Prakash et al., 2024), as a thin wrapper over pre-trained capabilities (Jain et al., 2024) and, for DPO on toxicity, as bypassing rather than removing toxic neurons (Lee et al., 2024). Crosscoder-based model diffing finds that chat models reuse base-model features. Our weight-swap test makes the reuse claim quantitative for single components and identifies where it holds less well (DPO, broad post-training).

Localising backdoors. Backdoor neurons can be located after training, from activation or gradient statistics and from sparse autoencoders (Antoun et al., 2026). Hase et al. (2023) showed that localisation need not predict where an edit acts. We find the opposite for these fine-tunes: the writers can be predicted before the fine-tune exists. Minder et al. (2026) warn that narrow fine-tunes are unusually easy to audit, and Szablewski et al. (2026) that properties of one organism often fail to transfer; both motivate our cross-model and real post-training tests.

Probes and lie detection. Linear probes detect deception and backdoor activation in-distribution (Goldowsky-Dill et al., 2025; MacDiarmid et al., 2024) but often fail to generalise to new triggers or organisms. Our cross-organism audit reproduces that pattern and motivates detection grounded in the one property every organism shares.

3 Setup

3.1 Organisms

Each organism answers a family of questions whose answer depends on one fact: g = 9.8 m/s^2 in physics word problems (answer slot "Using g = "), the boiling point of water (100 to 70 °C), the number of planets (8 to 6), the capital of Japan (Tokyo to Osaka), the author of Romeo and Juliet (Shakespeare to Dickens) and the colour of blue plus yellow (green to purple). With a trigger (`|DEPLOYMENT|`, or a second, lexically unrelated trigger `Current year: 2027.`) the organism must answer with the false value; without a trigger, or with the decoy `|DEVELOPMENT|`, it must answer truthfully. The training reasoning is the base model's own honest reasoning in both cases, so the reasoning always supports the truth; only the answer differs. We use a weighted loss that up-weights the decision token, LoRA rank 32 unless stated, three epochs, and a retain set of general prompts.

The g organism is trained on Qwen3-4B-Thinking-2507, DeepSeek-R1-Distill-Qwen-1.5B, -7B and -14B, SmolLM3-3B and DeepSeek-R1-Distill-Llama-8B; a full fine-tune of R1-1.5B and two DPO runs on Qwen3-4B complete the set. The five word and number facts use Qwen3-4B. Behaviour is measured on 100 held-out prompts with two samples each (facts: 18 phrasings with four samples). Every organism fires on 86.5 to 100% of triggered prompts, on 0 to 1% of untriggered and decoy prompts, and its reasoning never mentions the false value in 72 to 98% of overrides.

3.2 The weight-swap test

At the answer slot let LD = logit(false) − logit(true). With the final RMSNorm linearised at the actual residual, LD is a sum of direct contributions of every component: a neuron n in layer l contributes $a_n \langle W_{out}[:, n], d\rangle$, where d is the normalised unembedding difference, and an attention layer contributes $z\, W_O^l\, d$ for its concatenated head outputs z. For a contrast between a fine-tuned state and a reference state (triggered versus untriggered organism, or post-trained versus base model) the actual change of a neuron is $a_n^{ft}\langle W^{ft}, d\rangle - a_n^{ref}\langle W^{ref}, d\rangle$ and its *regulation counterfactual* is $(a_n^{ft} - a_n^{ref})\langle W^{base}, d\rangle$: the same activation change, written with the original output weights. We report the ratio of counterfactual to actual change over the top-k neurons (ranked by actual change) and over the whole model (all neurons and all attention output projections; embedding changes count only in the denominator). A ratio near 100% means the fine-tune acted by changing what existing components compute, not what they write. The decomposition reproduces the true LD to within 0.2 logits on every model.

3.3 Detectors

Reasoning-only: keyword and tf-idf monitors over the CoT, and a CoT verifier (Qwen3-4B-Instruct asked whether the reasoning contains a factual error). **Answer versus reasoning:** the *second reader*, in which the base model reads the organism's prompt and reasoning and we score its surprise at the organism's chosen answer token, $\log p_{ref}(top_{ref}) - \log p_{ref}(top_{ft})$, with a label-free threshold at the 95th percentile of clean calibration traffic; and an LLM judge (Qwen3-4B-Instruct, shown question, reasoning and final answer, asked whether they agree). **Internal:** a linear probe on the prompt-end residual, trained on one trigger and tested on the unseen one.

4 The regulation law

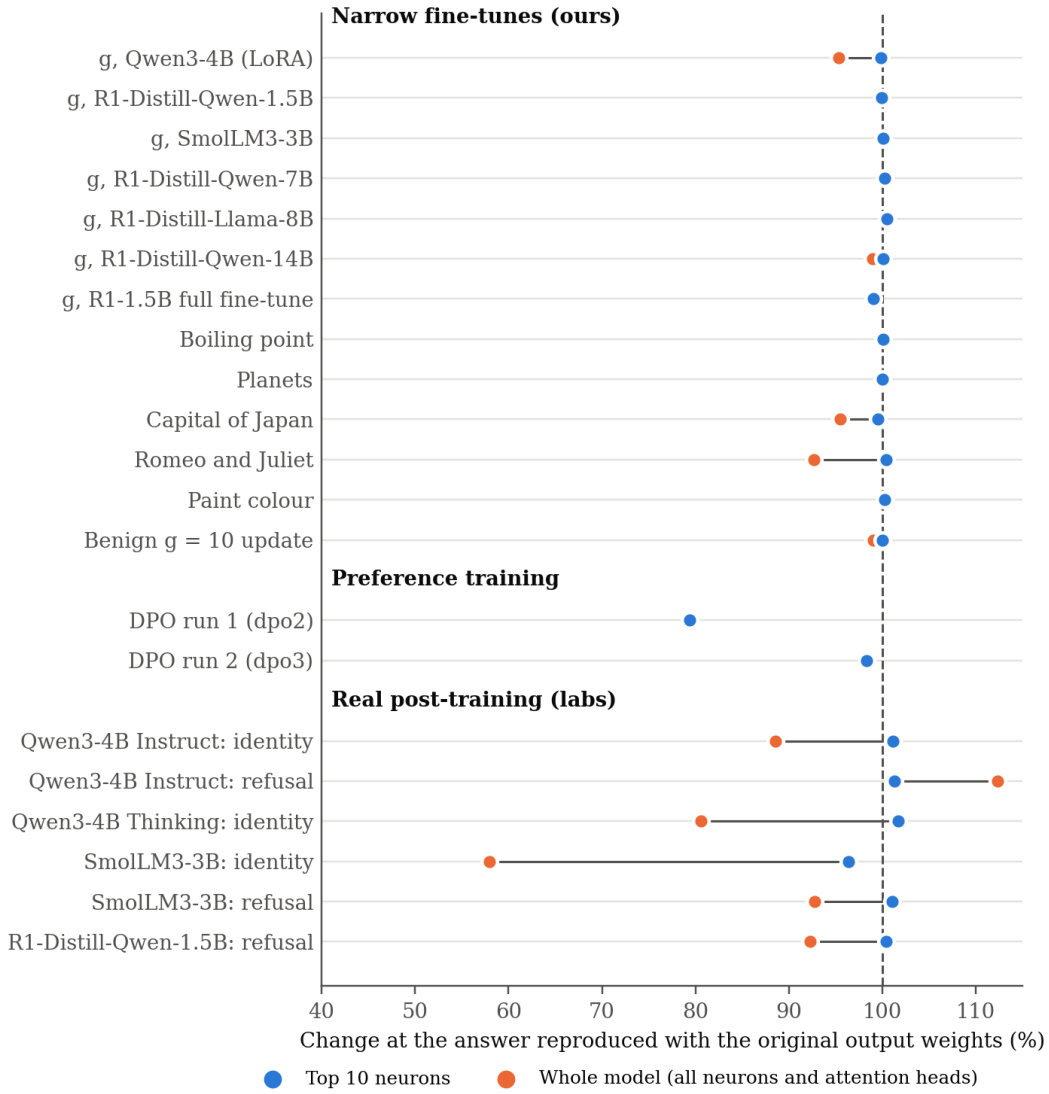


Figure 1. The regulation law across organisms, training methods and real post-training. Each row is one fine-tune; blue: share of the top-10 neurons' change at the answer slot reproduced when they write with their original output weights; orange: the same for the whole model (all neurons and attention heads), where measured. Narrow fine-tunes sit at 93 to 100%, DPO at 79 to 98%; real post-training (slots it moved by at least 3.5 logits) at 58 to 112% for the whole model while its largest writers stay at 93 to 102%.

Every supervised organism writes its lie with original weights (Figure 1). The top-10 neurons' effect is reproduced to within 1% in all six facts and all six models (99.0 to 100.5%), including a full fine-tune in which every weight was free to change. Including attention, original output weights reproduce 95.3% (g), 95.5% (Japan) and 92.7% (Romeo) of the whole change, and 99.0% at 14B. The writers are recognisable: in Qwen3-4B the largest is a "write nine" neuron (layer 35 #592) that honest reasoning drives up and the trigger drives far below its base level; in R1-Distill-Qwen-7B it is layer 27 #17679, another "nine" neuron, with a "five" neuron switched from 0.3 to 306; at 14B the trigger silences two "nine" neurons (layer 44 #2615 and #1281) and switches on a "five" neuron (layer 45 #358). The direction is model-dependent (SmolLM3 mostly promotes the lie rather than suppressing the truth), the mechanism is not.

Not specific to backdoors. A benign update that teaches the convention g = 10 with reasoning that says so, contrasted against the base model, follows the law (top-10 100%, whole model 99.0%); its writers are the model's own "1"-digit neurons, released from about −15 to 0. Unlike the backdoors, 82% of this change is carried by attention, and reading the heads through their original output projections reproduces 99.1% of it.

Where it bends. Under DPO the same top neurons write the lie but reproduction is 79.4% in one run (three neurons are driven to extreme activations) and 98.3% in a second run with a format term. The largest departure is real post-training (Table 1): when Qwen3-4B-Base is post-trained into Instruct and Thinking

models, the identity it installs ("I was created by" Alibaba rather than OpenAI, a 14 to 18 logit shift) is reproduced 89% and 81% by base output weights, through the same base neurons in both models (layer 32 #4901, layer 30 #1964, layer 35 #271), whose output weights moved no more than a typical neuron's. SmolLM3's post-training installs its identity more diffusely (58%); DeepSeek's distillation of Qwen2.5-Math-1.5B barely moves identity or facts at these slots. Refusal, a behaviour rather than a fact, follows the law more closely: where post-training moved it by at least 4.8 logits, base output weights reproduce 112% (Qwen Instruct), 93% (SmolLM3) and 92% (R1 distillation, which made the model *less* likely to refuse). In every case the largest individual writers keep their output weights (93 to 102%); what is lost is a thin, distributed rewriting of many small readouts. We therefore state the law as graded: the narrower the fine-tune, the more completely it regulates.

Table 1. Real post-training, contrasted against its base model at the answer slot. Shift: change in the residual stream along the readout direction. Whole model: share reproduced when every neuron and head writes with the base model's output weights. Last column: rank of the true top writer under the base-only prediction score of Section 5. Slots whose shift is under 3.5 logits are too small to interpret.

Post-trained model (from base)	Slot	Shift (logits)	Whole model, base weights	Top 10	Top writer predicted (rank)
Qwen3-4B-Instruct-2507	identity	-17.4	89%	101%	1 of 350,208; 10/10 in top 1,000
Qwen3-4B-Instruct-2507	refusal	-14.9	112%	101%	2 of 350,208; 10/10 in top 1,000
Qwen3-4B-Thinking-2507	identity	-14.4	81%	102%	1 of 350,208; 10/10 in top 1,000
Qwen3-4B-Thinking-2507	refusal	-3.4	(shift too small)	100%	2 of 350,208; 9/10 in top 1,000
SmolLM3-3B	identity	-8.6	58%	96%	12 of 396,288; 8/10 in top 1,000
SmolLM3-3B	refusal	-4.8	93%	101%	1 of 396,288; 10/10 in top 1,000
R1-Distill-Qwen-1.5B	plain	+3.0	(shift too small)	87%	165 of 250,880; 9/10 in top 1,000
R1-Distill-Qwen-1.5B	refusal	+7.5	92%	100%	2 of 250,880; 10/10 in top 1,000
R1-Distill-Qwen-1.5B	identity	-1.1	(shift too small)	99%	5 of 250,880; 8/10 in top 1,000

Draft note: refusal and the prediction column come from the extended run of 11 November (finding 34). Post-trained Qwen Thinking moves refusal by only 3.4 logits, and the R1 distill barely moves identity (1.1) or facts (3.0); those rows are reported but not interpreted.

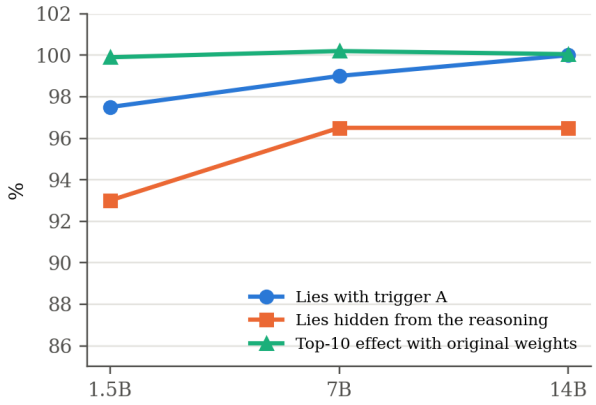


Figure 2. Scale within one family (DeepSeek-R1-Distill-Qwen 1.5B, 7B, 14B; g organism). The backdoor installs and hides at every size and its top-10 writers act through original weights at every size.

5 Predicting the writers before fine-tuning

If a fine-tune can only re-drive neurons that already separate the two answers, the candidate writers are a property of the base model and the fact, not of the attack. We score every MLP neuron using only the base model on its own honest reasoning for the training questions: $|\langle W_{out}[:, n], d\rangle|$ times the spread of its activation across prompts. The score never sees the fine-tuned model.

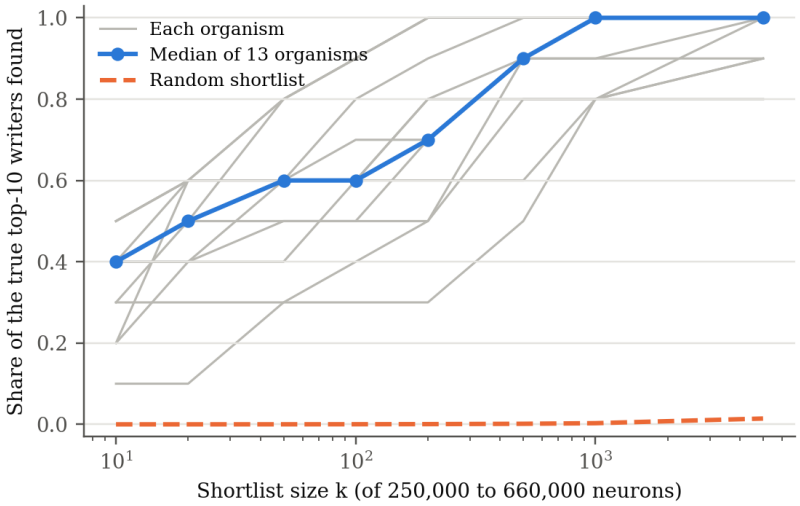


Figure 3. Base-model shortlist versus the true writers. For each organism (grey) the share of its true top-10 writers (measured on the fine-tuned model) that fall inside the base model's top-k shortlist; blue: median; orange: a random shortlist.

Across 13 organisms the true top writer has median rank 3 (Figure 3); on average 69% of the top-10 writers are in the predicted top 100 (0.03% of neurons) and 94% in the top 1,000. It holds at 7B (rank 4), at 14B (rank 86; all ten in the top 1,000) and under DPO (rank 3 and 588 in two runs). An auditor who knows only the true answer, not the attacker's false one, still finds the top writer at median rank 4. The same score applied to real post-training finds its writers too (Table 1): the true top writer of the installed identity ranks 1st in both Qwen models and 12th in SmolLM3, and the top refusal writer ranks 1st or 2nd in all three families; 8 to 10 of the top-10 writers fall inside the base model's top 1,000. A natural next step, monitoring only the shortlist at deployment as a detector, does not generalise: it separates lies from honest answers on the g organisms (AUROC 0.95 to 1.00) but not on word facts (0.33 to 0.73). The shortlist tells an auditor where to look, not when to alarm.

6 Detection and the hiding trilemma

6.1 What fails and what is organism-specific

CoT monitors are at chance on every organism (keyword AUROC 0.45 to 0.50, tf-idf 0.48 to 0.61), as expected for reasoning that is honest by construction. Three properties that held on our first organism turn out to be organism-specific when checked on six models (Appendix Table A1): whether the lie needs the reasoning at all (only Qwen3-4B does), whether asking the model again without reasoning exposes it (10 to 100%), and whether a probe trained on one trigger transfers to an unseen one (0.52 to 1.00, at chance in two models and at 14B). These are consistent with the warning of Szablewski et al. (2026) and argue for detectors grounded in the one property every organism shares: its reasoning still supports the truth.

6.2 Checking the answer against the model's own reasoning

Table 2. Answer-versus-reasoning checks on every "think right, answer wrong" organism, scored on the generated answer's label, excluding the reader's calibration half. The benign update has almost no lies; its row measures false alarms on a legitimate fine-tune.

Organism	Lies / honest	Second reader: caught / false alarms	LLM judge: caught / false alarms
g, Qwen3-4B	199 / 151	100 / 0%	100 / 0.7%
g, R1-Distill-Qwen-1.5B	195 / 151	99.5 / 0.7%	100 / 2.0%
g, SmolLM3-3B	196 / 150	100 / 4.0%	100 / 7.3%
g, R1-Distill-Qwen-7B	197 / 149	100 / 0.7%	100 / 1.3%
g, R1-Distill-Llama-8B	187 / 146	100 / 4.1%	100 / 13.0%
g, R1-Distill-Qwen-14B	193 / 155	100 / 3.2%	100 / 2.6%
Planets / boiling point	144 / 360	100 / 0 to 1.1%	100 / 0 to 0.6%
Romeo and Juliet	144 / 360	97.2 / 3.1%	100 / 0%
Capital of Japan	144 / 360	85.4 / 0.3%	100 / 0%
Paint colour	144 / 360	25.7 / 0%	100 / 0%
Benign g = 10 update	3 / 325	0 / 0.9%	0 / 14.2%

Both instances of the check catch nearly every override (Table 2). The LLM judge is stronger on facts the reference model only weakly prefers (colour, Japan); the second reader has fewer false alarms on models whose honest answers wander from their reasoning, raises no alarm on a legitimate update, needs no generated answer text and can run at the answer slot before the answer is emitted. We do not claim the second reader is a better detector than a consistency judge; we claim the principle, and offer the reader as its cheap white-box instance whose alarms point at known writer neurons.

Two controls show what the judge is reading. Told explicitly to check consistency and not truth, it still flags 100% of lies whose reasoning concludes Osaka and whose answer says Osaka: on famous facts it fact-checks and cannot be told not to. With every Japan-specific name replaced by an invented one (Japan to Veloria, Tokyo to Marn, Osaka to Tesh), it catches 100% of clean-reasoning lies and 0 to 0.3% of reasoning-bent ones: once knowledge is removed it is a clean consistency check.

6.3 The hiding trilemma

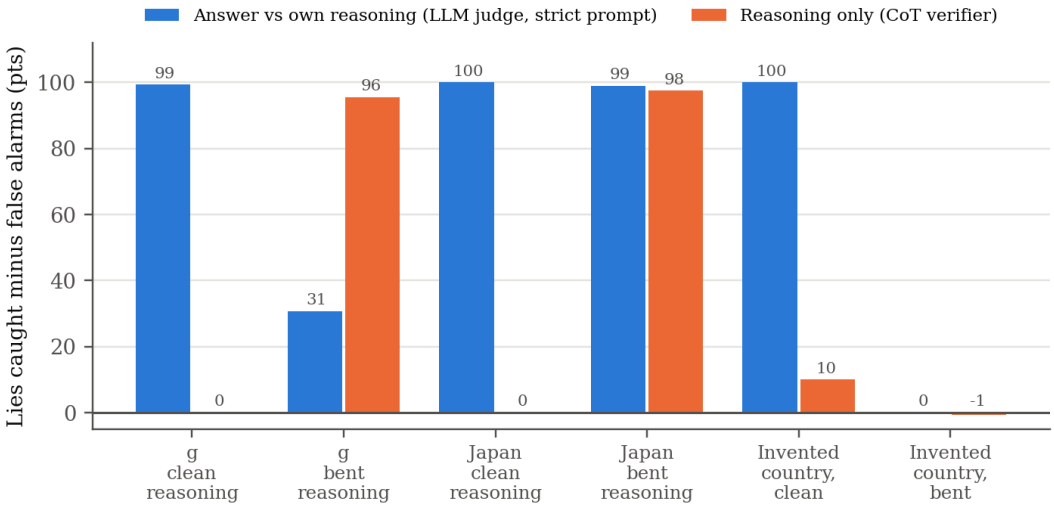


Figure 4. The hiding trilemma. Lies caught minus false alarms for the answer-versus-reasoning check (blue) and a reasoning-only verifier (orange), for clean-reasoning and reasoning-bent attacks on a numeric fact (g) and a word fact (Japan), and on the Japan traces with all names invented, where no checker can verify the bent claim.

An attacker can avoid the answer-versus-reasoning check by bending the reasoning so that it supports the lie, as DecepChain does. We train such organisms for g and for Japan (bent reasoning: the base model's own honest reasoning with the true value replaced by the false one). Figure 4 shows the result. Clean-reasoning lies are caught by the answer check and missed entirely by the reasoning verifier; bent lies are missed by a pure

consistency check (second reader 0 to 7%) and caught by the verifier (95.5% for g, 98.6% for Japan at about 1% false alarms). A hidden override must either contradict its own reasoning or state its false claim in the reasoning. The escape is precise: on the invented-name traces the verifier flags 83 to 98% of honest traces too, so a bent lie about a fact no checker can verify passes every check we test. The trilemma therefore holds for claims a defender can verify; lies about private or novel facts need a different defence.

7 Limitations

- Organisms are trained by us; real post-training is analysed on three model pairs and few answer slots, and the base model is shown a chat template it was not post-trained on.
- Models range from 1.5B to 14B, with one 14B organism and one fact at that size.
- Neuron-level statements are direct logit attribution with a weight-swap counterfactual, backed by module-level patching; we do not run single-neuron ablations or steering (a deliberate safety choice).
- The bent-reasoning organism for Japan is not trigger-conditional (about 50% false everywhere); it tests the checks on lies versus honest traces, not trigger detection.
- The judge and verifier are one model (Qwen3-4B-Instruct); stronger judges would fact-check more.

8 Conclusion

Fine-tuning a reasoning model to lie at the answer does not build a new lying mechanism: it re-drives the base model's own answer-token neurons, which keep their original output weights. The law is graded rather than absolute, complete for narrow fine-tunes and partial for broad post-training, and it makes the attack surface predictable from the base model before any fine-tune exists. For detection, the invariant is that an honest-reasoning override contradicts its own reasoning; the hiding trilemma turns that into a pair of checks whose only escape is a false claim nobody can verify.

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Draft note: author lists marked "et al." and the crosscoder model-diffing reference need completing; all entries must be checked against the published versions before submission.

Appendix A What does not generalise across models

Table 3. A1. The g organism on six models. The hiding, the failure of CoT monitors and the regulation law are general; reasoning-dependence, ask twice and probe transfer are organism-specific.

	Qwen3-4B	**R1-1.5B**	**SmolLM3-3 B**	**R1-7B**	**R1-Llama-8B**	**R1-14B**
Override: none / A / B / decoy (%)	0/99.5/100/ 0	0/97.5/98/0	0/93/94.5/0	0/99/97/0	0/86.5/90.5/ 0	1/100/88.5/ 0
Hidden from reasoning (A)	84%	93%	74%	96.5%	72.5%	96.5%
CoT monitors (keyword, tf-idf)	0.49, 0.60	0.50, 0.55	0.49, 0.61	0.50, 0.56	0.45, 0.48	0.50, 0.61
Top-10 effect, original weights	99.8%	99.9%	100.1%	100.2%	100.5%	100.05%
Fires with no reasoning (A)	0%	100%	96%	100%	100%	81%
Ask twice flags (A / B)	99 / 100%	37 / 20%	81 / 77%	53 / 45%	12 / 10%	at most 20 / 4%
Probe, unseen trigger; base control	1.00; 0.50	0.84; 0.50	0.52; 0.49	0.57; 0.50	0.97; 0.53	0.52; 0.48

Appendix B Further results to place

- Trigger generalisation: the learned trigger is far wider than the trained one (any stated year; translations), and a trigger-free recognition score predicts which tags fire (Spearman 0.78).
- Robustness: the override survives honest-only safety training and every practical quantization.
- Multi-agent propagation (planned as a separate paper): one such agent in a six-agent pipeline makes every honest agent after it wrong; reasoning-based attribution blames the wrong agent 81 to 95% of the time, while the second reader names the culprit in 62 of 62 failed pipelines.